Lecture 10: Classification metrics

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Machine Learning II: 46-927

December 2, 2020

Announcements

- Homework was just due. There will be a quiz on Friday. Everything is back to the usual schedule.

Today

Today we will talk more about classification metrics, and then potentially introduce generative models.

- Talk about the homework
- Classification:
  - Imbalanced data
  - Classification metrics
- Maybe a bit of decision trees

Classification so far

Focus on binary classification,

- $Y = 1$ if an event of interest happened
- $Y = 0$ if it did not happen

Most of our classifiers will give us a guess $\hat{p}(x) = \hat{\mathbb{P}}(Y = 1|X = x)$.
This is an estimate of the true $p(x) = \mathbb{P}(Y = 1|X = x)$.

We saw that to minimize average misclassification error,
$\mathbb{E}\mathbf{1}_{f(X)\neq Y}$, you should choose $f(x) = \mathbf{1}_{p(x)>1/2}$.

Let's think about this more carefully.

Misclassification rate

Suppose that I tell you my classifier has 97% accuracy. Is this good?

Misclassification rate

Suppose that I tell you my classifier has 97% accuracy. Is this good?

It depends! What if I am predicting lottery wins. The probability of winning may be $1/175000000$.

In that case, guessing "no" has a misclassification rate of 5×10^{-9}...

We need to consider the *base rate*: the probability of each class given no further information.

Confusion matrix: Example

For the homework problem (to be), $Y = 1$ if an individual responds to marketing. The classification rule $\hat{p}(x) > 0.5$ gives a confusion matrix similar to:

	True $Y = 1$	True $Y = 0$
Guess $Y = 1$	0	2
Guess $Y = 0$	1083	7957

Issues with imbalanced data and misclassification rates

The idea of minimizing the misclassification rate is tempting — no one wants to be wrong!

However, this is often not the real goal of applying a classifier. Losses are usually asymmetric! Sometimes you even care directly about $\hat{p}(x)$!

Furthermore, in settings where the classes are imbalanced, the misclassification rate is very difficult to improve and is often misleading.

Imbalanced data

In many common applications, our classes are imbalanced:

- Marketing
- Fraud detection
- Mortgage default
- Whether there will be a large jump in price after an event.

Misclassification rate and imbalanced data

We know that we should classify as $Y = 1$ when $P(Y = 1|X = x) > 0.5$ to minimize expected misclassification rate. Suppose that an event only happens 5% of the time in the general population.

By Bayes Theorem,

$$P(Y = 1|X = x) =$$

Therefore, to classify as $Y = 1$, we would need to see an X that is more likely in the $Y = 1$ population than the general population.

Even if there is useful information in the data, it is probably not enough to flip our classifications!

Confusion matrix: error rates

	True $Y = 1$	True $Y = 0$
Guess $Y = 1$	True Positive	False Positive
Guess $Y = 0$	False Negative	True Negative

Other error rate measures better capture asymmetric concerns when classifying:

- Sensitivity (Recall, Hit rate):

- Specificity (True Negative Rate, TNR):

These quantities will allow us to better think about cost trade offs.

Sensitivity and Specificity

Sensitivity: Fraction of points with $Y = 1$ that we find.

- Want high Sensitivity when False Negatives are more costly than False Positives. Example: Fraud detection.

Specificity: Fraction of points with $Y = 0$ that we avoid.

- Want high Specificity when False Positives are more expensive than False Negatives. Example: Criminal trials.

We would like both of them to be high, but we need to make tradeoffs as we adjust our threshold.

- We can flag more cases as fraud to improve sensitivity, but this will decrease our specificity.

Cost/Loss for classification

	True $Y = 1$	True $Y = 0$
Guess $Y = 1$	0	L_{01}
Guess $Y = 0$	L_{10}	0

You will show on your homework that the loss is minimized if you guess $Y = 1$ when

$$\mathbb{P}(Y = 1|X = x) >$$

If it costs \$5 dollars to call someone who will not buy, but it costs \$100 to miss someone who would buy, you should

This suggests that we can trade off costs and error rates by just changing the threshold on $\mathbb{P}(Y = 1|X = x)$!

Changing threshold on $\mathbb{P}(Y = 1|X = x)$

As we change the threshold on $\mathbb{P}(Y = 1|X = x)$, we vary the size and quality of the set we reject.

We can think about $\mathbb{P}(Y = 1|X = x)$ as a function for ranking our observations by likelihood of $Y = 1$.

It is convenient to look at the Sensitivity and Specificity of our classifier as the threshold changes. This corresponds to the ROC (Receiver operating characteristic) curve that you will plot in your homework.

Aside: Classification scores

So far, we've been talking about classification based on estimates $\hat{p}(x) = \hat{\mathbb{P}}(Y = 1|X = x)$.

In fact, we don't really need probabilities to make classifications. A more general concept is a classification *score*, $\hat{f}(x)$, such that $\hat{f}(x)$ is big for

Any threshold on $\hat{f}(x)$ will yield a confusion matrix, which in turn yields all of the error metrics we have discussed, as well as ROC curves.

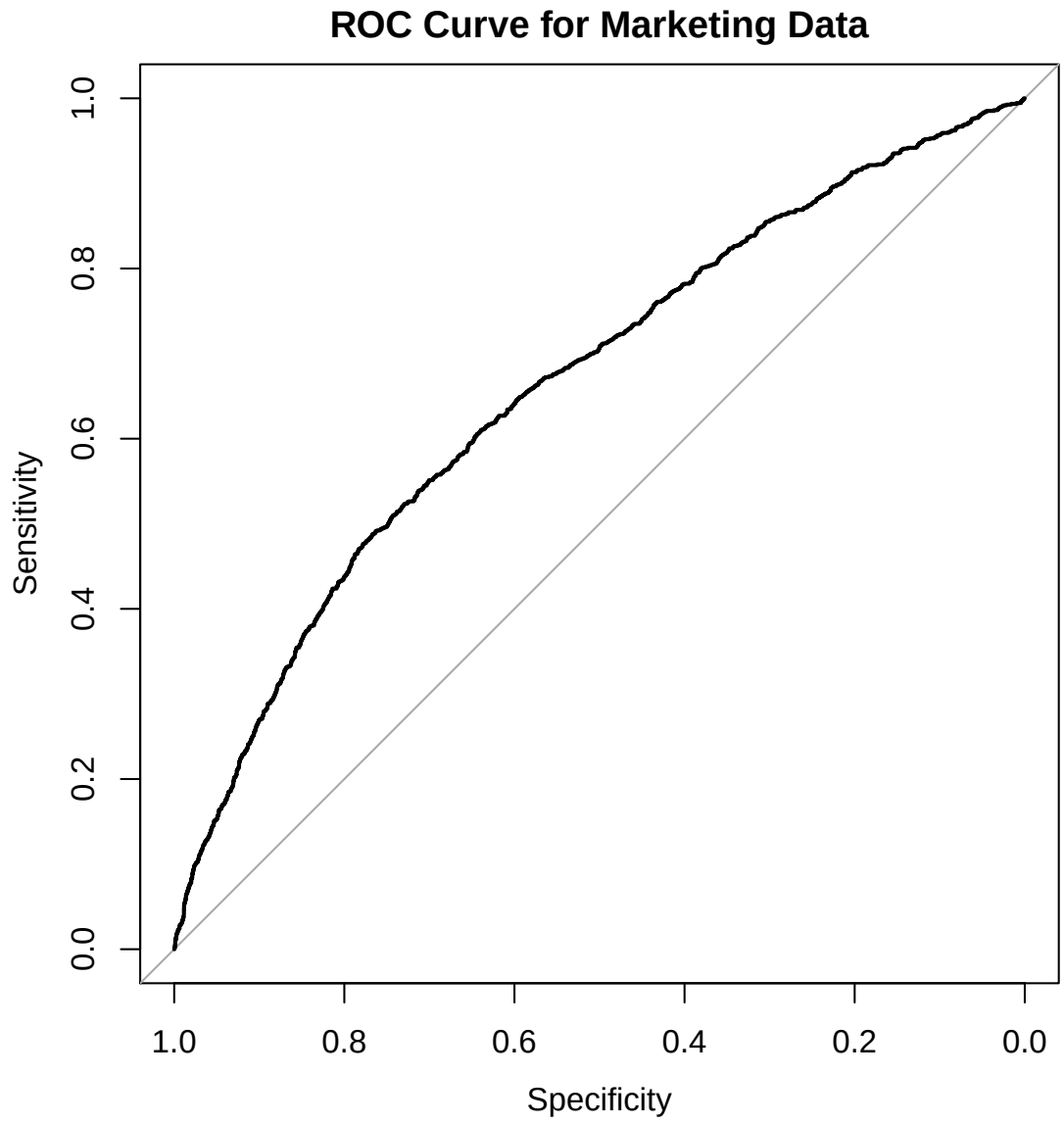
- Question: How do ROC curves change if I monotonically warp my classification score?

Aside: Classification scores

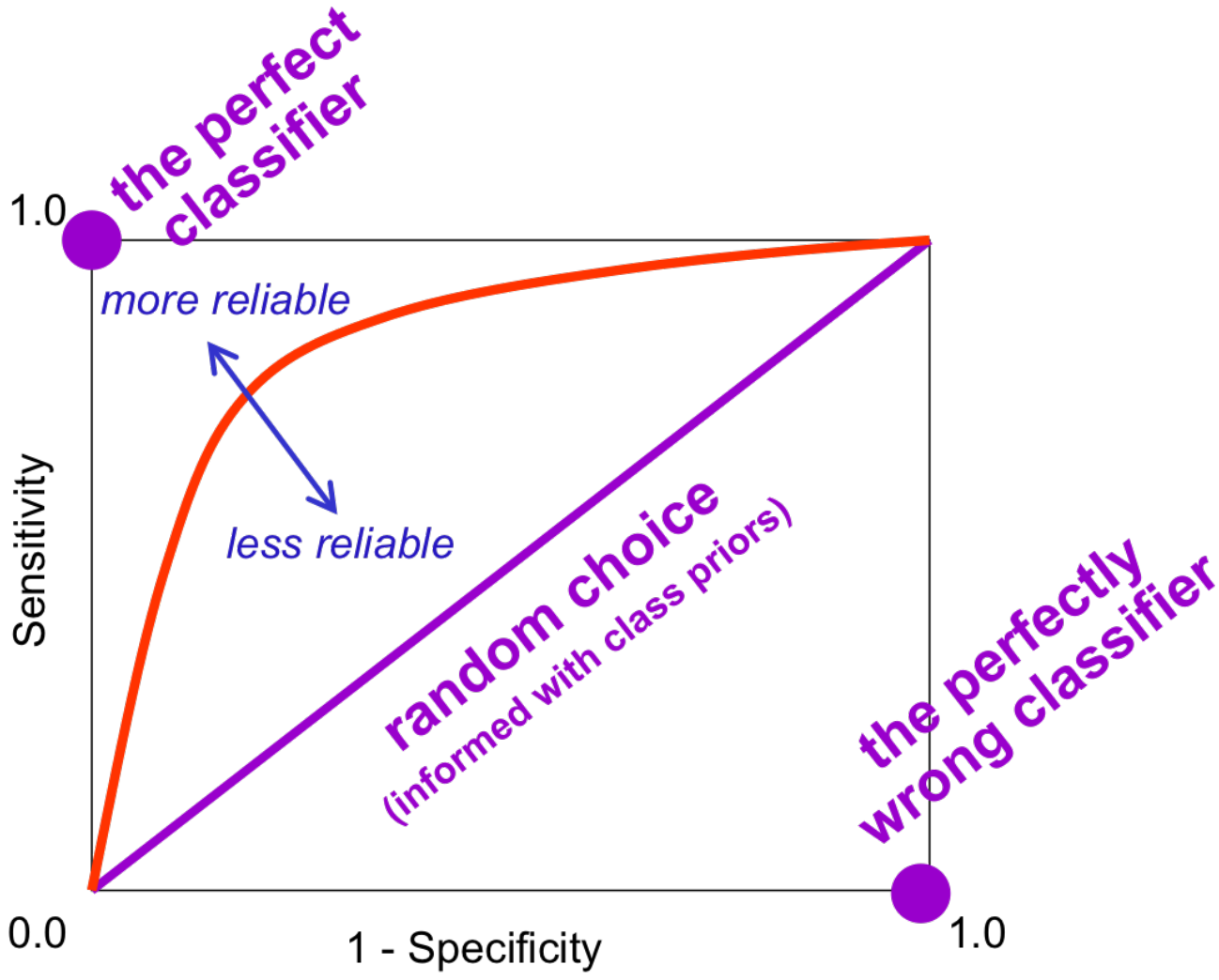
Examples of classification scores that are not probabilities:

It is often useful to think of the distribution of $\hat{f}(x)|Y = k$ for understanding classifier behavior. (Homework)

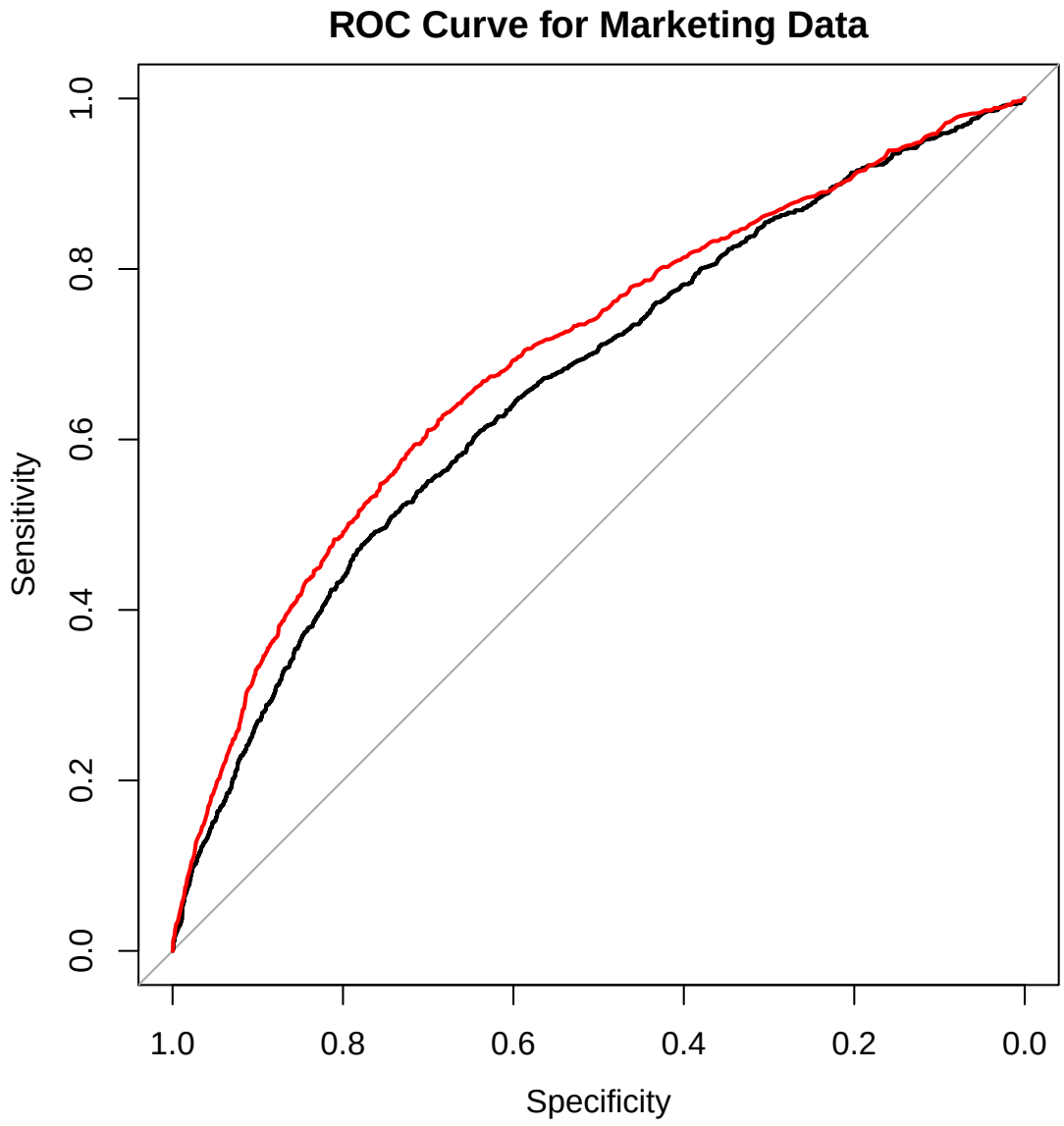
ROC Curve


ROC Curve for Marketing Data
Sensitivity
0.0
0.2
0.4
0.6
0.8
1.0
1.0
0.8
0.6
0.4
0.2
0.0
Specificity

ROC Curve


the perfect classifier
1.0
more reliable
less reliable
Sensitivity
random choice
(informed with class priors)
the perfectly wrong classifier
0.0
1 - Specificity
1.0

ROC Curve (again)



ROC Curve

A generally higher curve tends to be better, though you should keep tradeoffs in mind.

The Area Under the Curve (AUC) is one way to measure this. A higher area tends to be more attractive.

AUC has a nice interpretation:

If you randomly pick two observations from your distribution and order them by $\hat{p}(x)$, the AUC is the probability that your ordering is correct!

In reality though, you are interested in the performance cutoff at a particular point on the curve

Other metrics

Positive predictive value (PPV):

$$PPV = \frac{\#\text{ observations correctly classified as positive}}{\#\text{observations classified as positive}}$$
$$= \frac{TP}{TP + FP}$$

Suppose that you take a test for a disease. The PPV captures the probability that you actually have the disease when you test positive.

Other metrics

Negative Predictive Value (NPV):

$$NPV = \frac{\#\text{ observations correctly classified as negative}}{\#\text{observations classified as negative}}$$
$$= \frac{TN}{TN + FN}$$

Suppose that you take a test for a disease. The NPV is the probability that you are not sick, given that the test comes up negative.

Calibration

Is our $\hat{p}(x)$ a good estimate of $p(x) = \mathbb{P}(Y = 1|X = x)$?

Why do we care? We're just going to threshold it...

There is information in knowing $\mathbb{P}(Y = 1|X = x)$.

- A case that is definitely fraud ($p(x) = 0.99$) may be handled very differently than a case that might be fraud ($p(x) = 0.52$).

  Calculating the proper trade-offs for making decisions depends on the probabilities, not just the categories.

Note: Even if $\hat{p}(x)$ is not a good approximation of $p(x)$, thresholding $\hat{p}(x)$ might still give good guess at categories.

Calibration plots

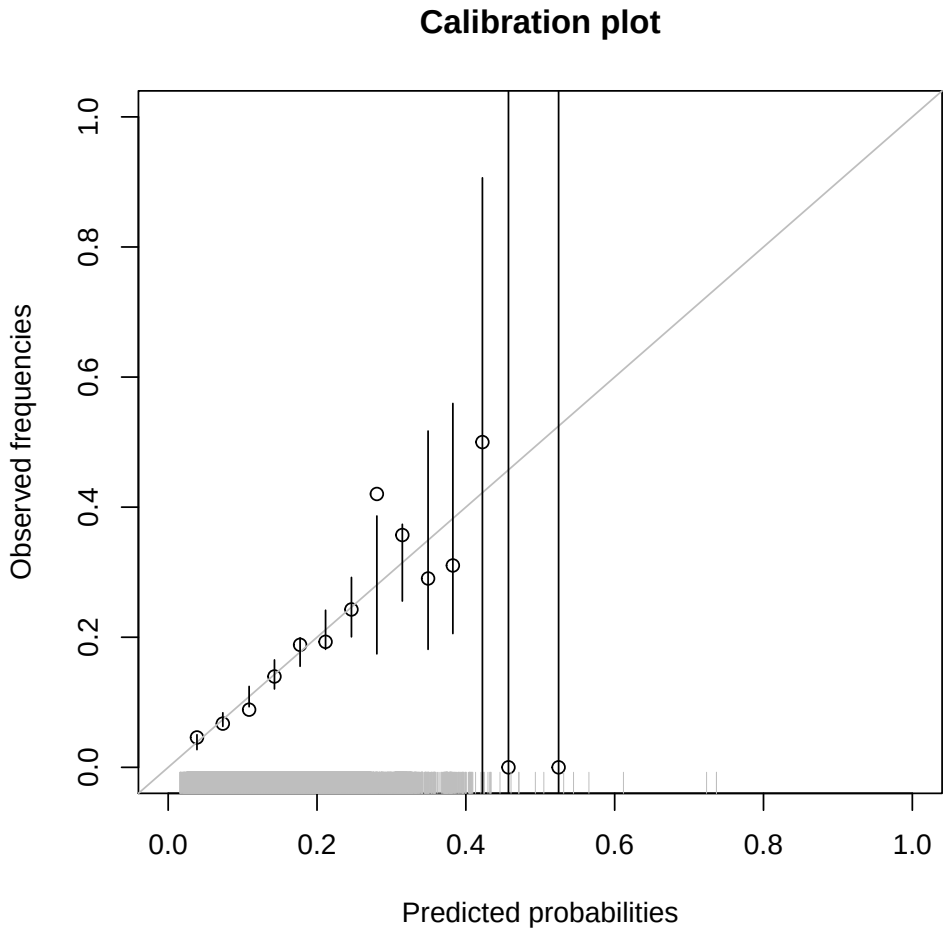
To check how well calibrated your $\hat{p}(x)$ is, you can make a *calibration plot*:

1. Bin the data according to predicted probability, $\hat{p}(x)$. E.g., $[0, 0.1], (0.1, 0.2], \ldots, (0.9, 1]$.

2. For each bin, calculate the proportion of observations with class $Y = 1$

3. Plot these true fractions against the midpoints of the bin predicted probabilities

Calibration plots



Calibration plots

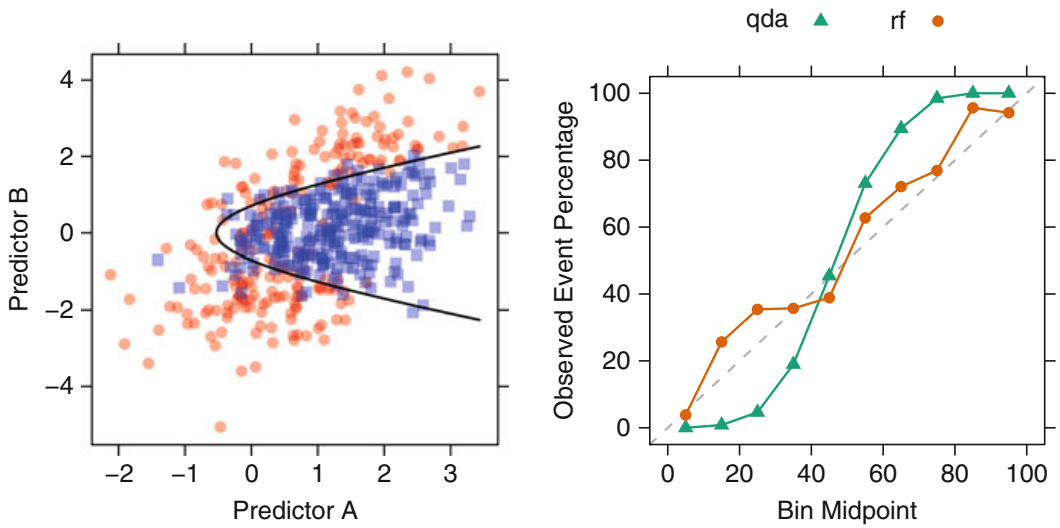


Fig. 11.1: *Left*: A simulated two-class data set with two predictors. The *solid black line* denotes the 50 % probability contour. *Right*: A calibration plot of the test set probabilities for random forest and quadratic discriminant analysis models

(From *Applied Predictive Modeling* by Kjell Johnson and Max Kuhn)

If the calibration is off, you can try transforming your $\hat{p}(x)$.

Remark: Model validation and tuning

Suppose that we want to pick between different classification models. Or instead, that we need to choose λ in logistic lasso. How can we do this?

Where are we headed next?

We showed that the optimal misclassification error is achieved by either of the following ideal classifiers:

$$\begin{aligned}\hat{f}(x) &= \underset{j=1,\ldots K}{\operatorname{argmax}} \ \mathrm{P}(Y = j | X = x) \\ &= \underset{j=1,\ldots K}{\operatorname{argmax}} \ \mathrm{P}(X = x | Y = j) \cdot \pi_j\end{aligned}$$

1. **Discriminative models:** We can try to estimate $\mathbb{P}(Y = j|X = x)$ directly.
   - Continue here with more complex models

2. **Generative models:** We could try to estimate the distributions $\mathbb{P}(X = x|Y = j)$ for each class.
   - Deferring until later / next mini

Logistic regression

We introduced logistic regression as a first example of a discriminative classifier.

We model $Y_i \sim \text{Bernoulli}(P(Y = 1|X = x))$ with

$$\text{P}(Y = 1|X = x) = \frac{e^{\beta^T x}}{1 + e^{\beta^T x}} = \frac{1}{1 + e^{-\beta^T x}}$$

and then maximize the likelihood.

However, these modeling choices can be quite restrictive. Even if we expand beyond simple linear functions, we have to specify a good functional form and the important interactions.

We will now construct an alternative set of discriminative classifiers that will circumvent these problems.

Flexible models

We've seen a couple flexible models:

- Additive models: flexible about transformations, but require interactions to be specified.
- k-nearest neighbors: can be flexible and predict well, with enough data and a good distance. However, nearly impossible to interpret! No simplification, requires all training data to make a prediction. Struggles in high dimensions.
- SVMs: (In the future, ML 2) Can capture interactions or transformations indirectly by changing kernels, but the connections are not direct and only some can be captured.

Today we are going to introduce new styles of flexible methods that avoid many of these problems.

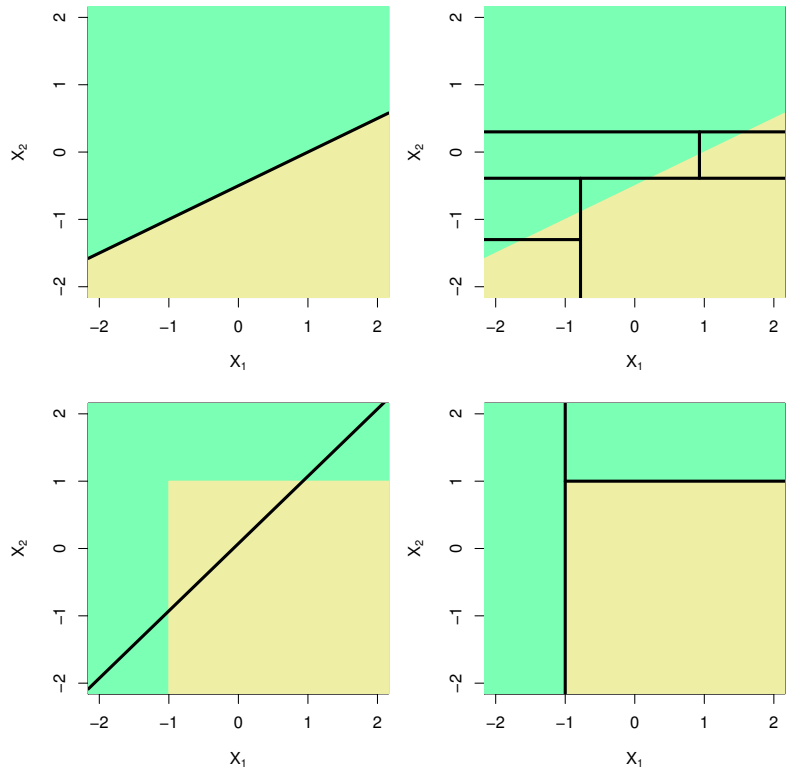
Wish list

There are many properties that we might want for a good prediction method.

- Good predictions and good probability estimates
- The ability to tune bias/variance
- Interpretability: What are our predictions based on?
- Automatic handling of variable transformations
- Automatic detection of interactions

Overview: Tree-based methods

Tree-based based methods for predicting y from a feature vector $x \in \mathbb{R}^p$ divide up the feature space into rectangles, and then fit a very simple model in each rectangle. This works both when y is discrete and continuous, i.e., both for classification and regression

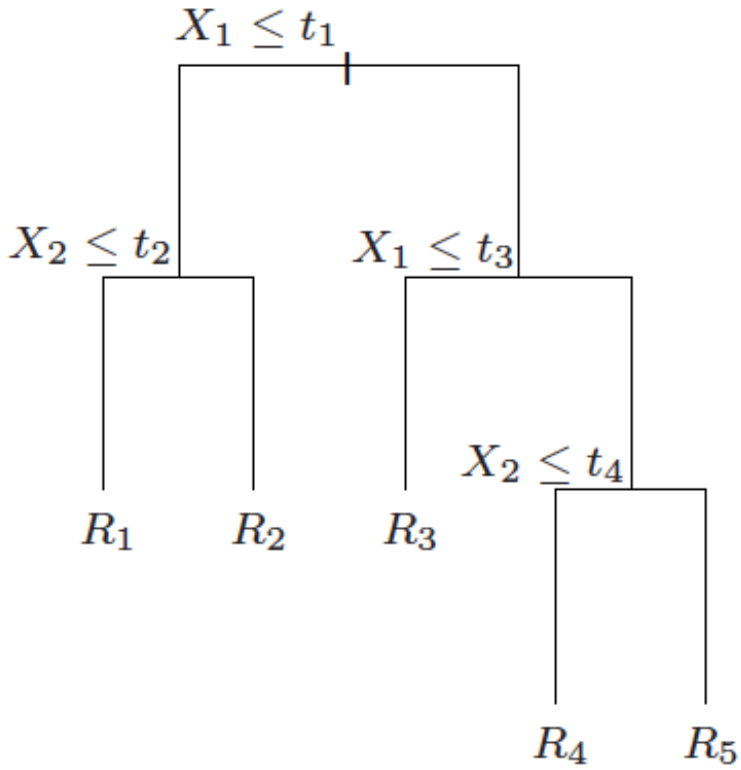


(ISL Figure 8.7)

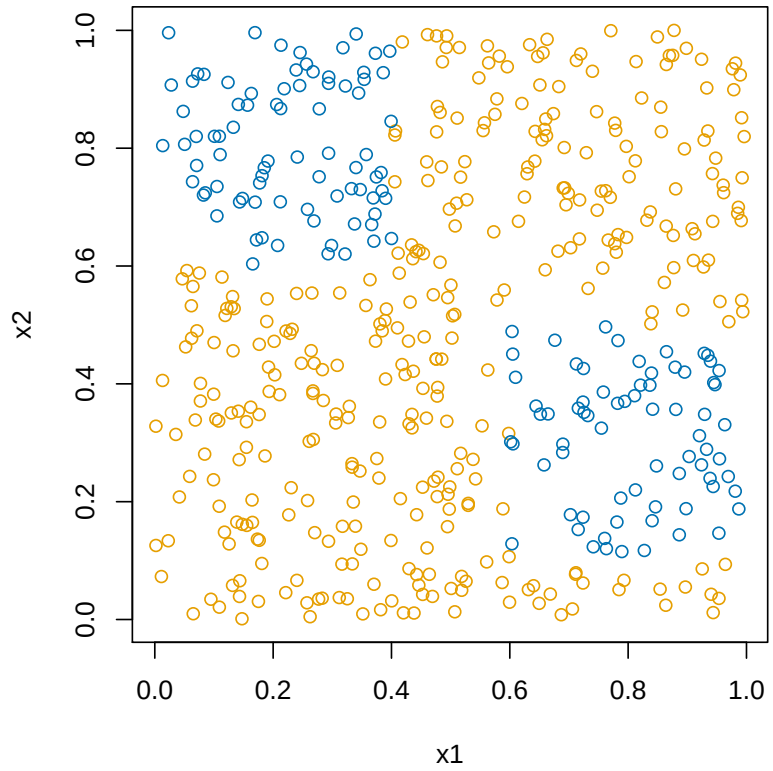
This is a big shift from thinking about linear-style models!

Tree-based methods

Tree-based methods will fix some of the problems we have encountered with other methods. They will give us a flexible rule, but one we can understand.



Overview: Tree-based methods

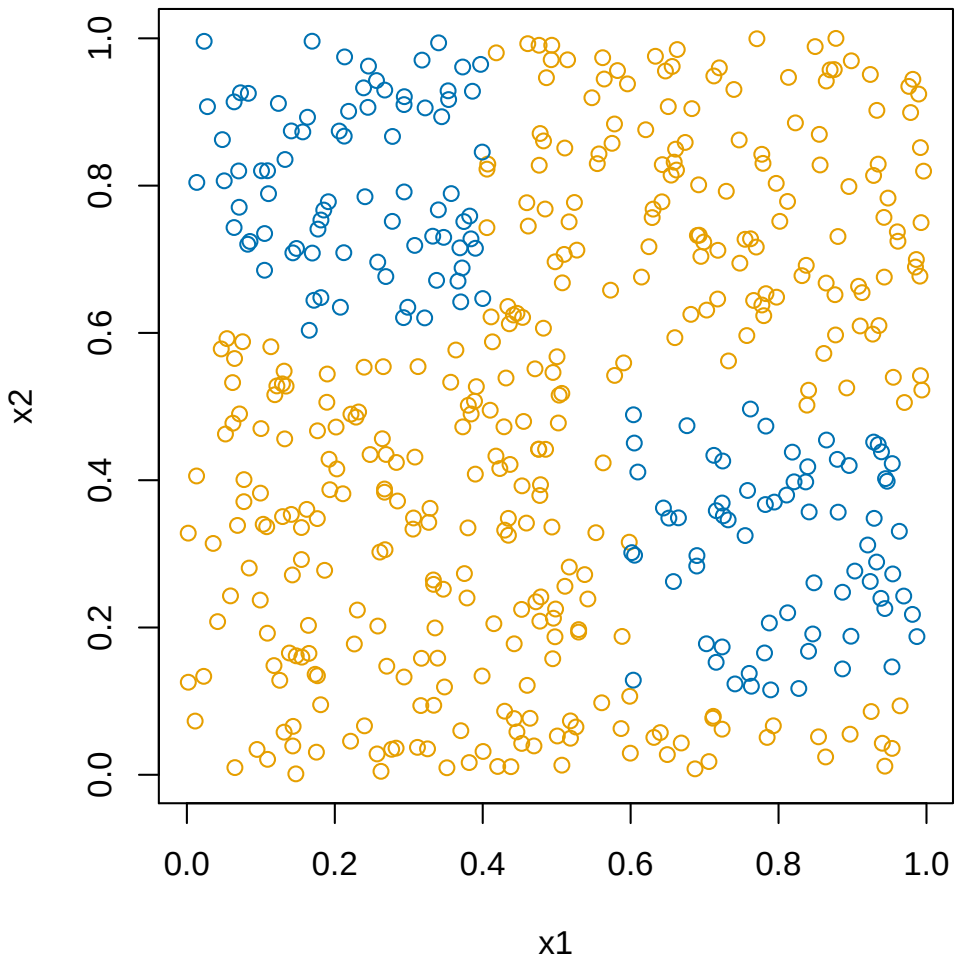


Does dividing up the feature space into rectangles look like it would work here?

Overview: Tree-based methods

Searching over all possible rectangular partitions is hard! There are many such partitions!

We make it simpler by considering a sequence of binary splits on individual variables.



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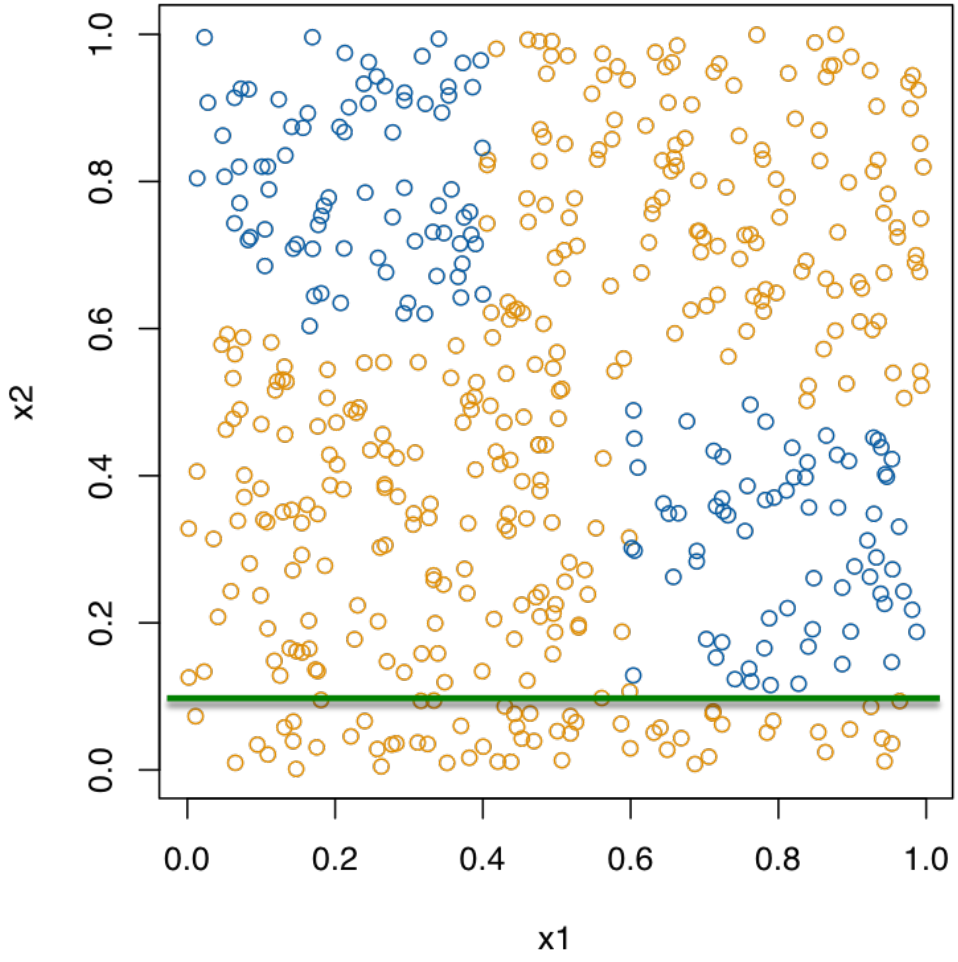

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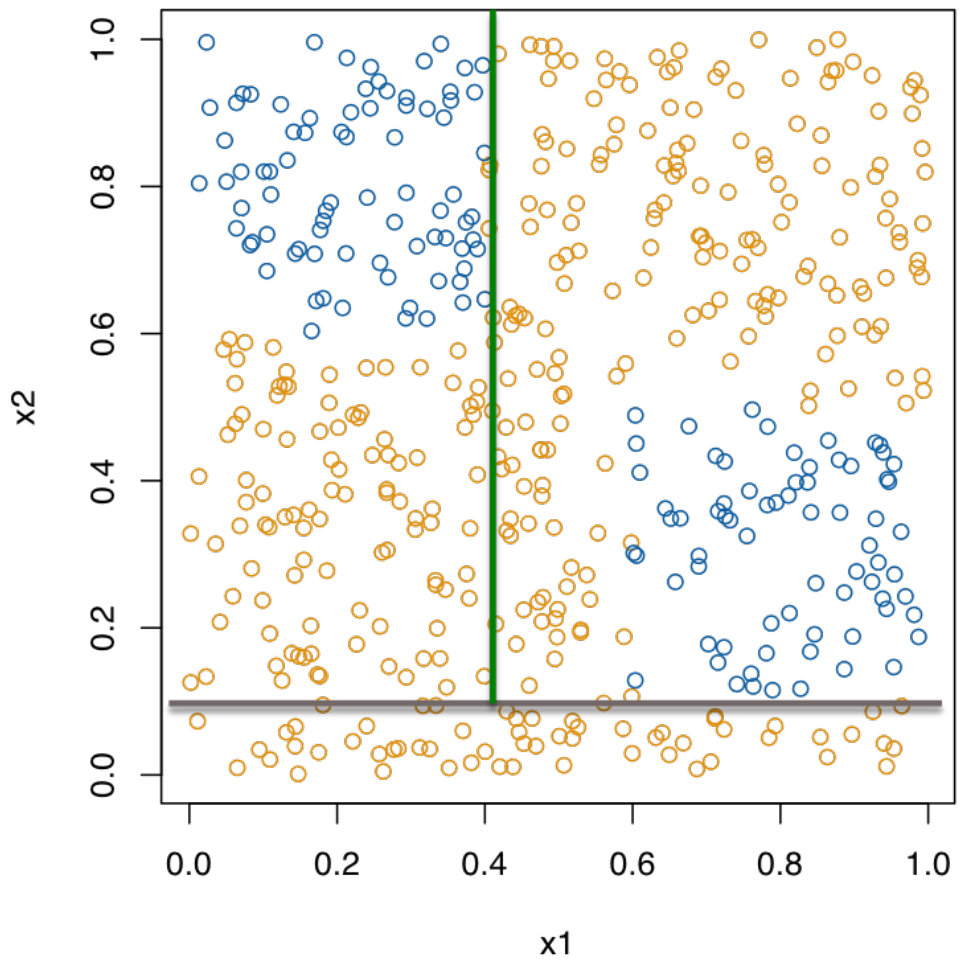

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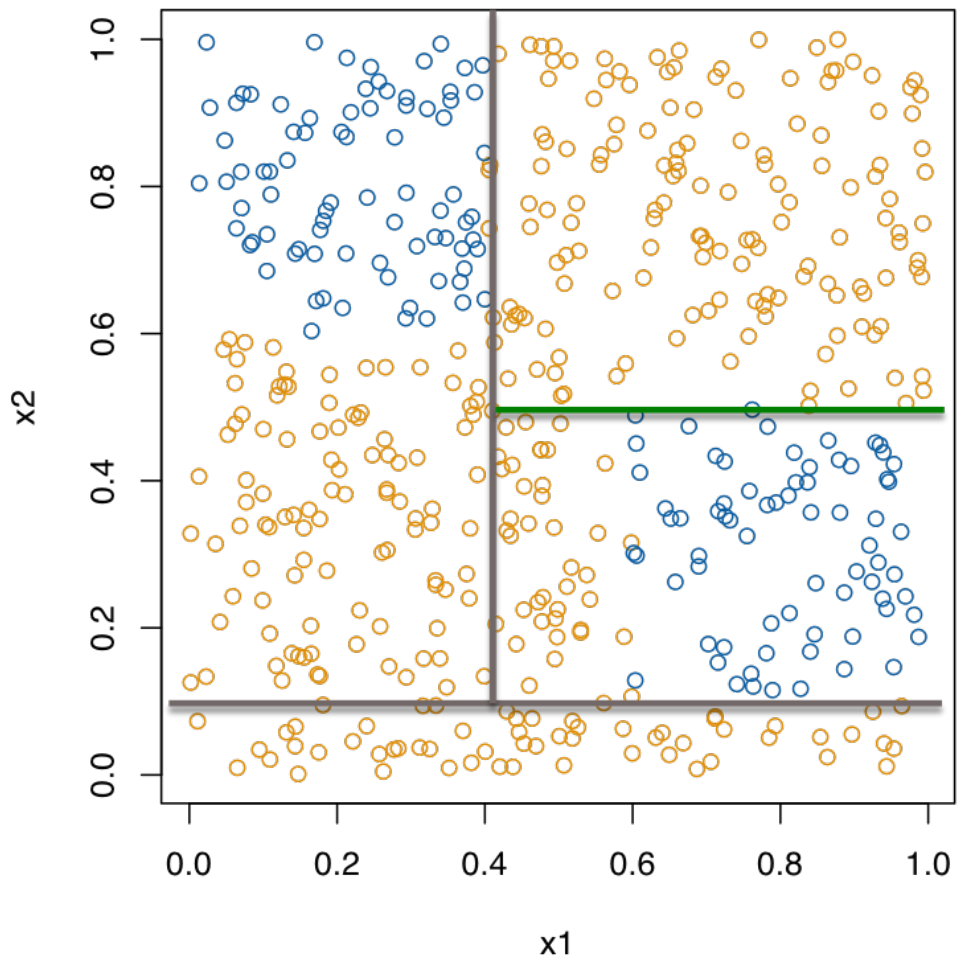

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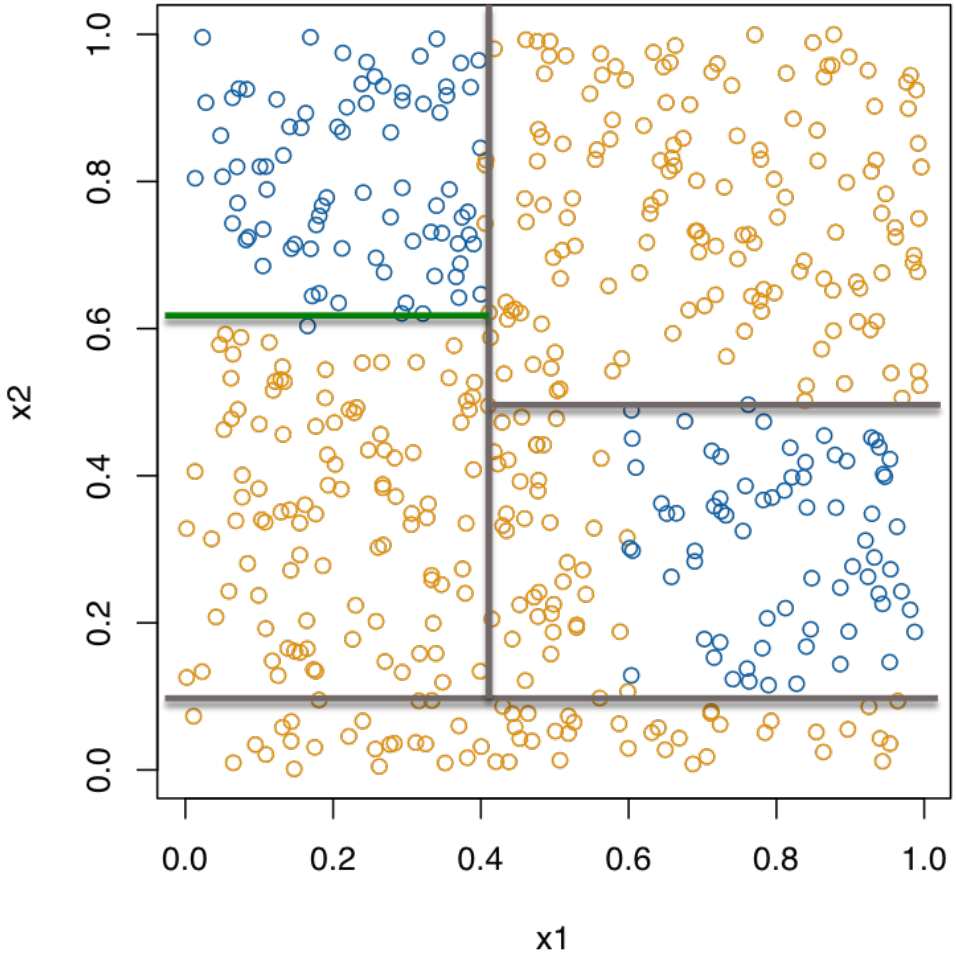

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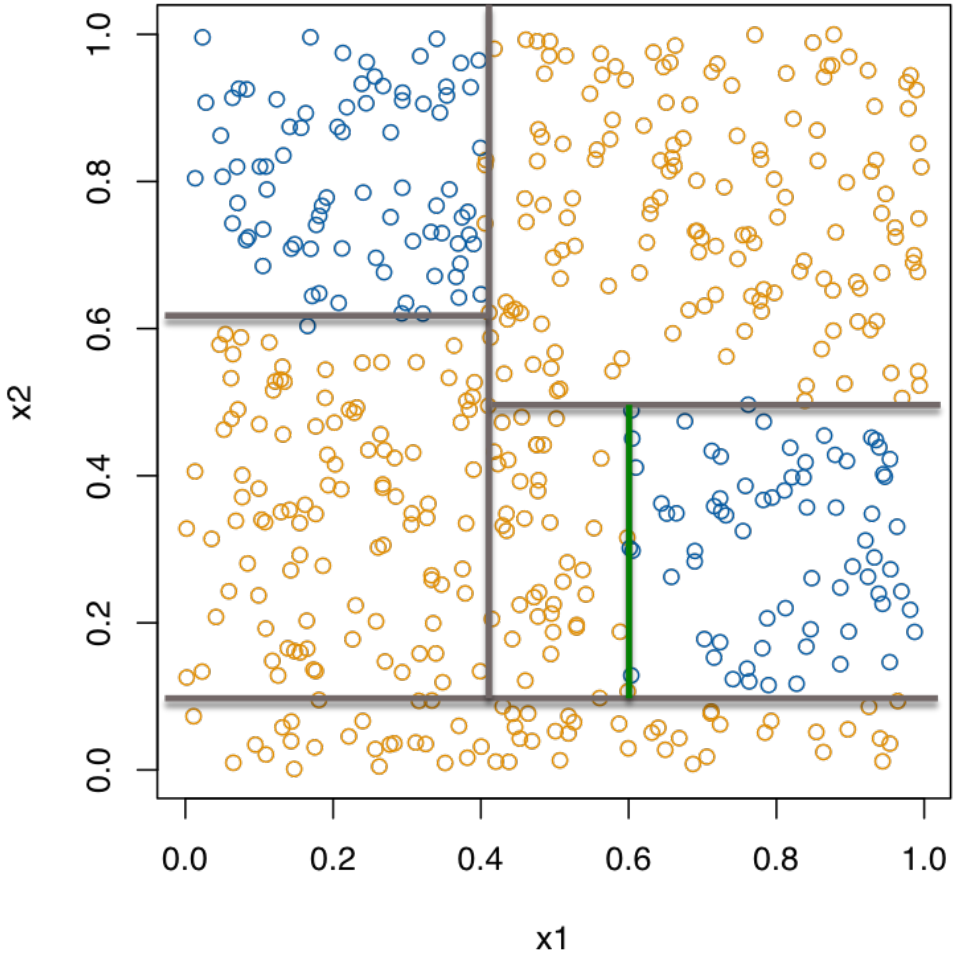


Overview: Tree-based methods

We simplify our partition search by only considering a sequence of binary splits on single variables.

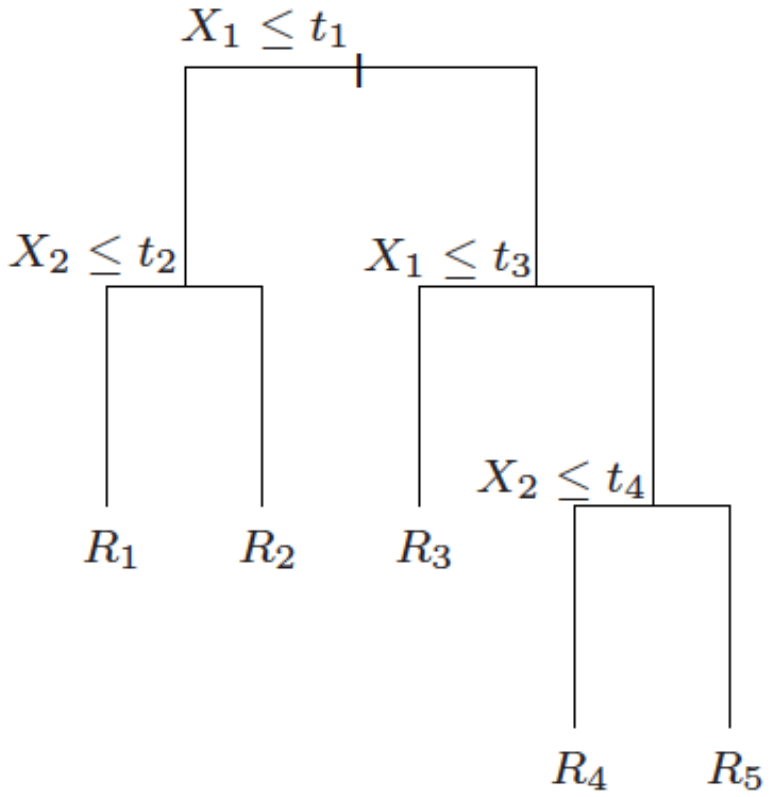
I.e., we choose a variable X_j, $j = 1, \ldots p$, divide up the feature space according to

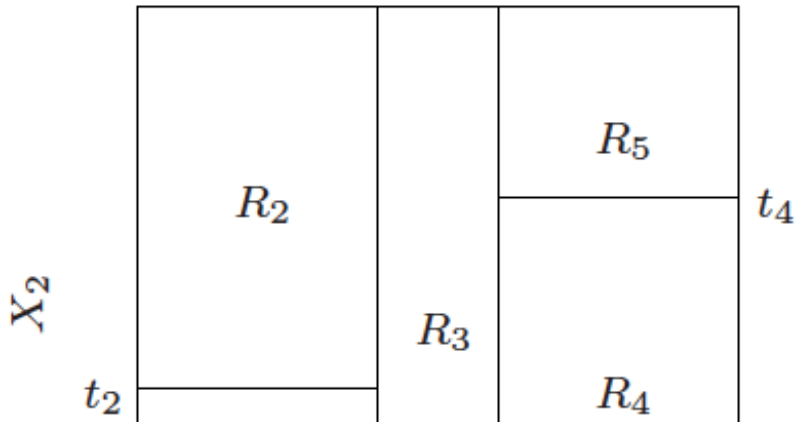
$$X_j \leq c \quad \text{and} \quad X_j > c.$$

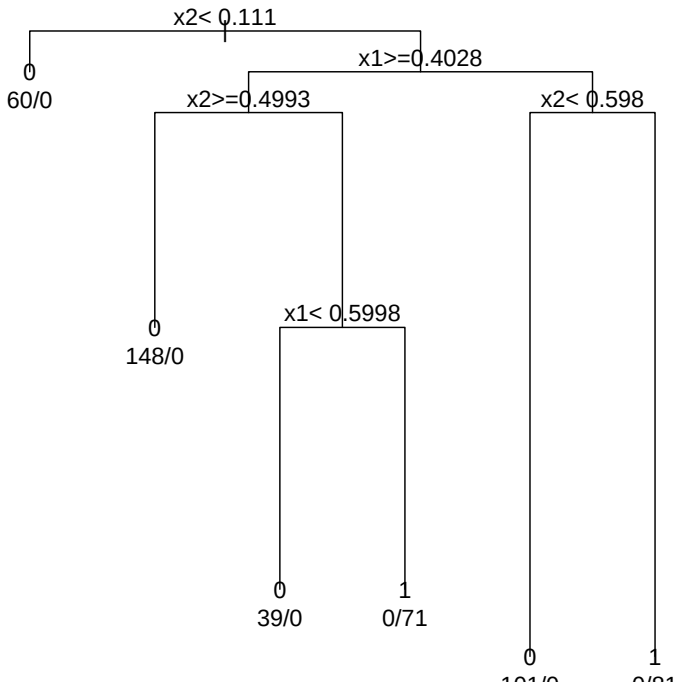
Then we proceed on each half separately.

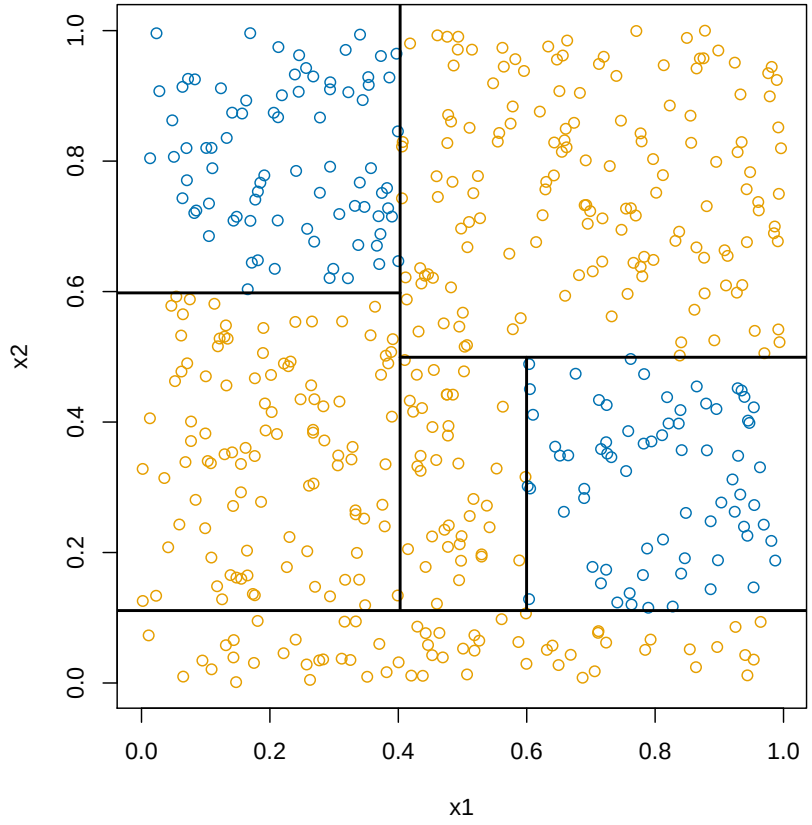
To make this even simpler, we choose the splits in a "greedy" fashion, only looking at the payoff for the current split and ignoring advantages for future splits.

A benefit of this search approach is that it also gives a natural tree representation of the partition.









This gives a rule that is easy to understand, easy to explain, and easy to implement!

No more coefficients to interpret!

Classification trees

Let's start with classification.

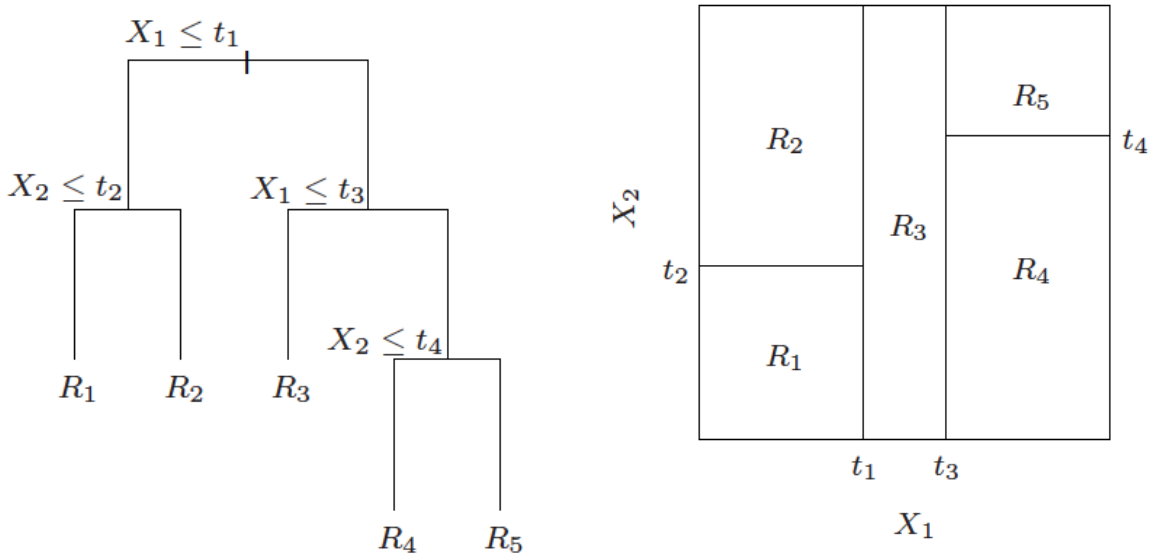
Classification trees are popular because they are interpretable, and sometimes because they mimic the way decisions are made.

Let (x_i, y_i), $i = 1, \ldots n$ be the training data, where $y_i \in \{1, \ldots K\}$ are the class labels, and $x_i \in \mathbb{R}^p$ measure the p predictor variables.

We are going to use a tree to define our classification function $\hat{f}(x) : \mathbb{R}^p \to \{1, \ldots, K\}$.

Think about the tree taking the place of our linear model in previous methods.

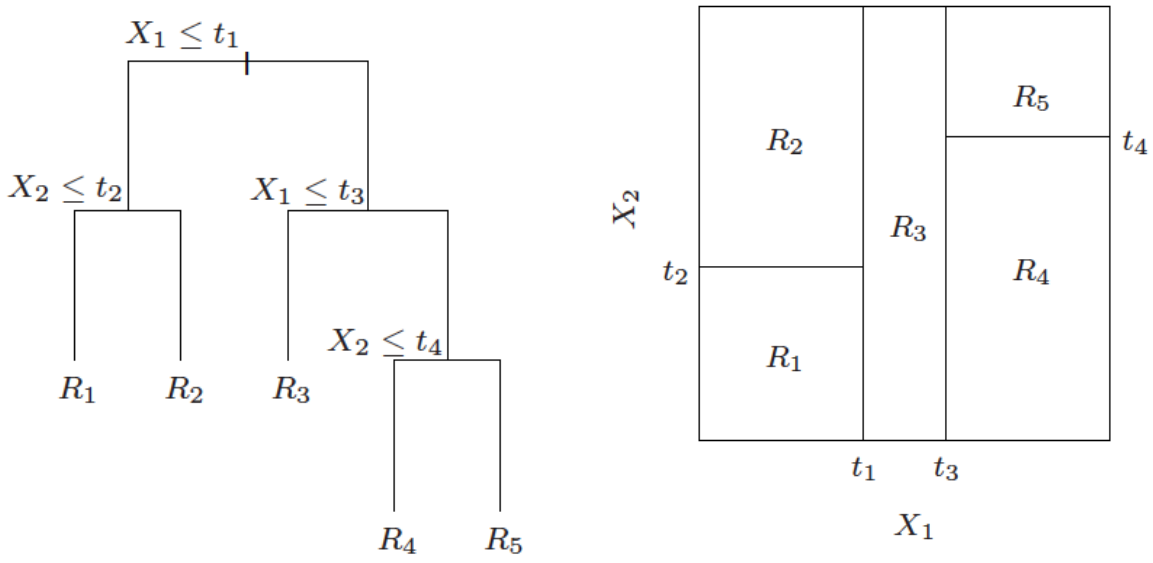
Classification trees



The classification tree can be thought of as defining m regions (rectangles) $R_1, \ldots R_m$, each corresponding to a leaf of the tree

We assign each R_j a class label $c_j \in \{1, \ldots K\}$. We then classify a new point $x \in \mathbb{R}^p$ by

$$\hat{f}^{\text{tree}}(x) \;=\; \sum_{j=1}^{m} c_j \cdot 1\{x \in R_j\} \;=\; c_j \;\text{ if }\; x \in R_j$$



$$\hat{f}^{\text{tree}}(x) = \sum_{j=1}^{m} c_j \cdot 1\{x \in R_j\}$$

Finding out which region a given point x belongs to is easy since the regions R_j are defined by a tree—we just scan down the tree. Otherwise, it would be a lot harder (need to look at each region)

Estimated class probabilities

We now have a flexible function that takes in covariates $x \in \mathbb{R}^p$ and produces classifications!

In the last few lectures, we realized that we want something more. We want actual estimates of the class probabilities!

How can we get these?

Estimated class probabilities

Note that each region R_j contains some subset of the training data (x_i, y_i), $i = 1, \ldots n$, say, n_j points.

We have been predicting class c_j using the most common class among points in R_j.

For each class k, we can also estimate the probability that a point has that class, given that it falls in R_j, $\mathrm{P}(C = k | X \in R_j)$, by

$$\hat{p}_k(R_j) = \frac{1}{n_j} \sum_{x_i \in R_j} 1\{y_i = k\}$$

the proportion of points in the region that are of class k.

We can even think of our predicted class as

$$c_j = \underset{k=1,\ldots K}{\operatorname{argmax}} \ \hat{p}_k(R_j)$$

How to build trees?

There are two main issues to consider in building a tree:

1. How to choose the splits?
2. How big to grow the tree?

Think first about varying the depth of the tree ... which is more complex/flexible, a big tree or a small tree? What tradeoff is at play here? How might we eventually consider choosing the depth?

Now for a fixed depth, consider choosing the splits. If the tree has depth d, it could have $\approx 2^d$ nodes. At each node we could choose any of p the variables for the split—this means that the number of possibilities is

$$p \cdot 2^d$$

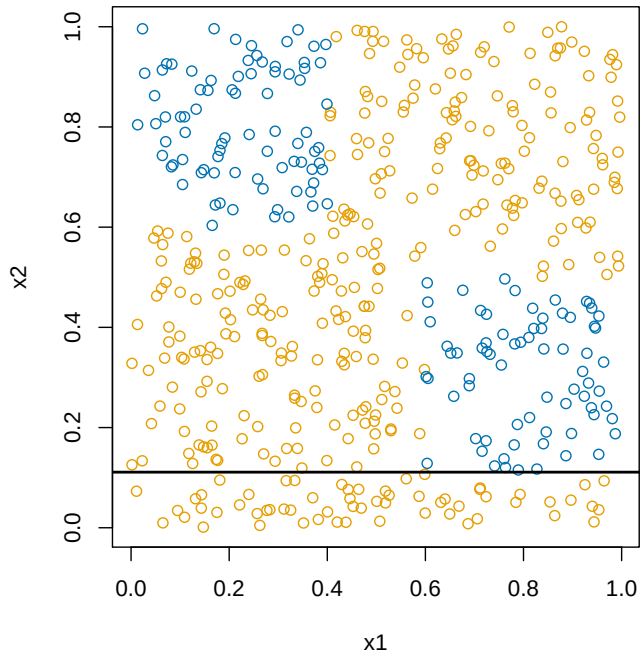
This is huge even for moderate d! And we haven't even counted the actual split points themselves

The CART algorithm

The CART algorithm[1] chooses the splits in a top down fashion: First chooses the first variable to be the root, then the variables at the second level, etc.

At each stage we choose the split to achieve the biggest drop in misclassification error—this is called a greedy strategy. In terms of tree depth, the strategy is to grow a large tree and then prune at the end.

Why grow a large tree and prune, instead of just stopping at some point?



[1] Breiman et al. (1984), "Classification and Regression Trees"

Recall that in a region R_m, the proportion of points in class k is

$$\hat{p}_k(R_m) = \frac{1}{n_m} \sum_{x_i \in R_m} 1\{y_i = k\}.$$

The CART algorithm begins by considering splitting on variable j and split point s, and defines the regions

$$R_1 = \{X \in \mathbb{R}^p : X_j \leq s\}, \quad R_2 = \{X \in \mathbb{R}^p : X_j > s\}$$

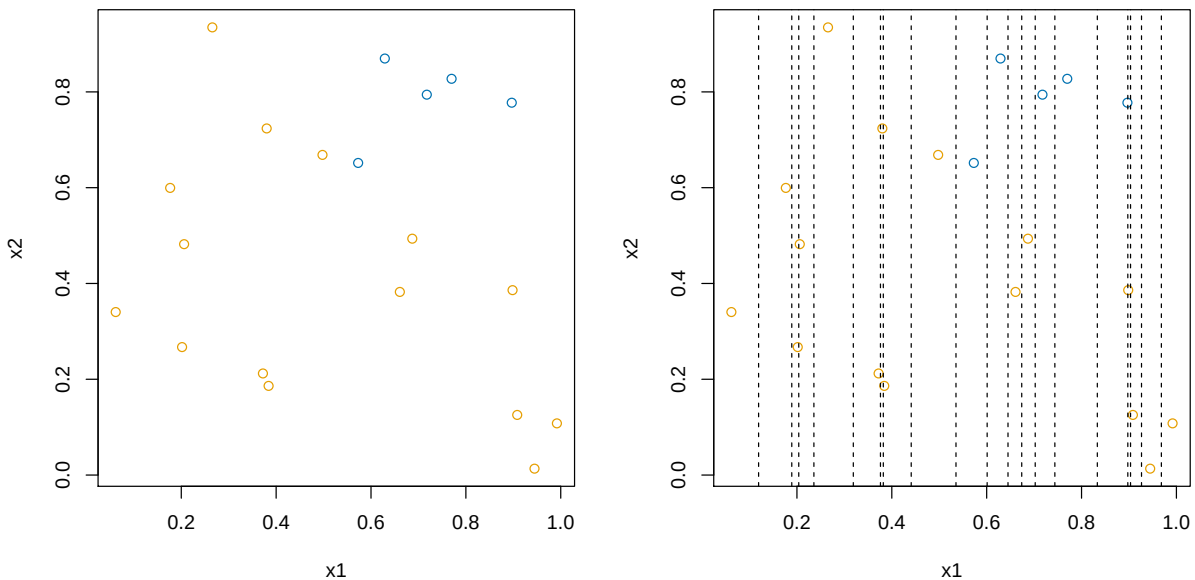
We then greedily chooses j, s by minimizing the misclassification error

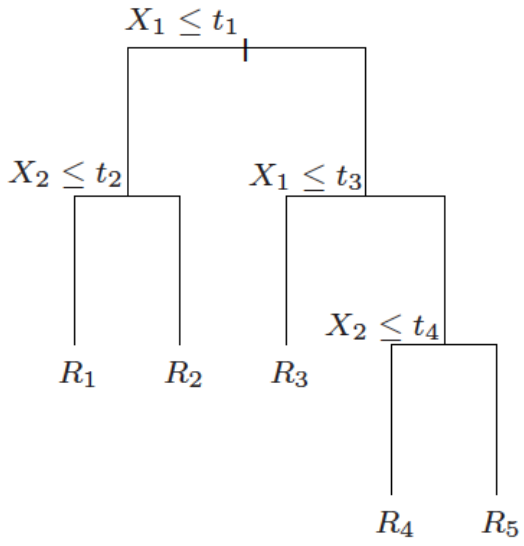
$$\underset{j,s}{\text{argmin}} \left(n_{R_1} \left[1 - \hat{p}_{c_1}(R_1)\right] + n_{R_2} \left[1 - \hat{p}_{c_2}(R_2)\right] \right)$$

Here $c_1 = \text{argmax}_{k=1,\ldots K}\ \hat{p}_k(R_1)$ is the most common class in R_1, and $c_2 = \text{argmax}_{k=1,\ldots K}\ \hat{p}_k(R_2)$ is the most common class in R_2

We now repeat this within each of the newly defined regions R_1, R_2. Again consider all variables and split points for each of R_1, R_2, greedily choosing the biggest improvement in misclassification error.

How
do we find the best split s? Aren't there infinitely many to consider?

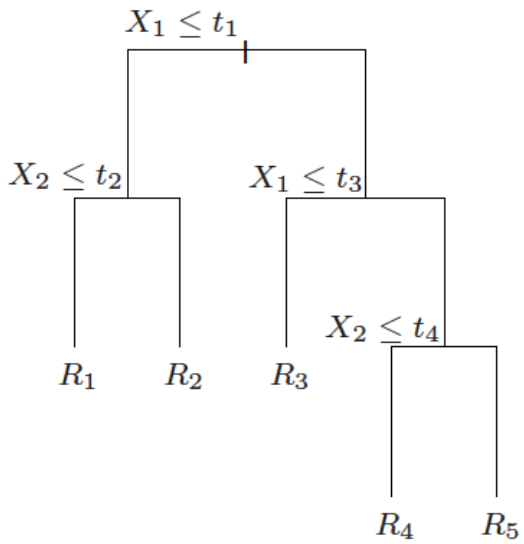




Continuing on in this manner, we will get a big tree T_0. Its leaves define regions $R_1, \ldots R_m$. We then prune this tree, meaning that we collapse some of its leaves into the parent nodes

How should we decide how much to prune a tree?

What is weird about applying this here?



For any tree T, let $|T|$ denote its number of leaves. We define

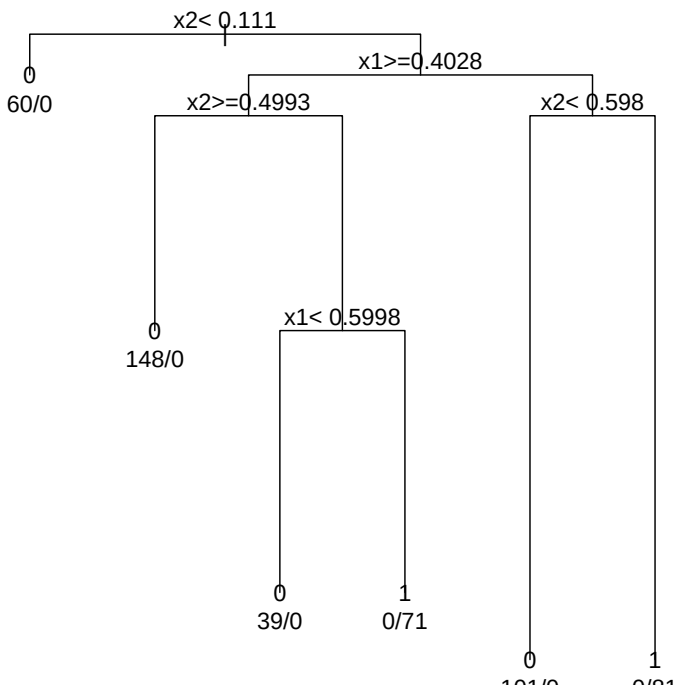
$$C_\alpha(T) = \sum_{j=1}^{|T|} \left[1 - \hat{p}_{c_j}(R_j)\right] + \alpha|T|$$

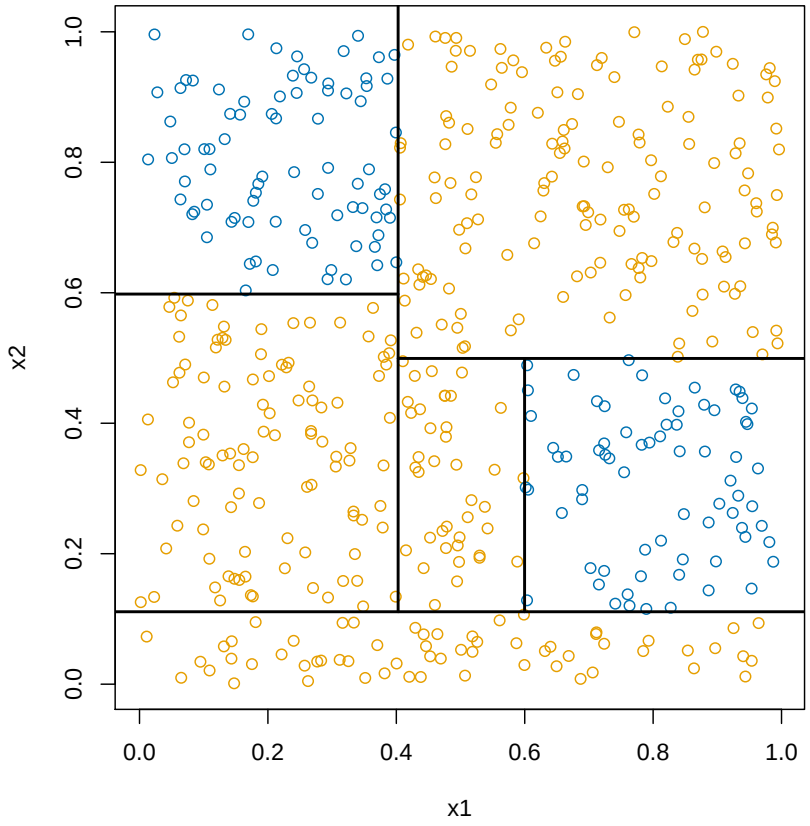
We seek the tree $T \subseteq T_0$ that minimizes $C_\alpha(T)$. It turns out that this can be done by pruning the weakest leaf one at a time.

Note that α is a tuning parameter, and a larger α yields a smaller tree. CART picks α by 5- or 10-fold cross-validation

Example: simple classification tree

Example: $n = 500$, $p = 2$, and $K = 2$. We ran CART:





In R, can use `rpart` or `tree`. In python, `DecisionTreeClassifer`. However, python does not yet support pruning (as far as I can tell)!!

Example: spam data

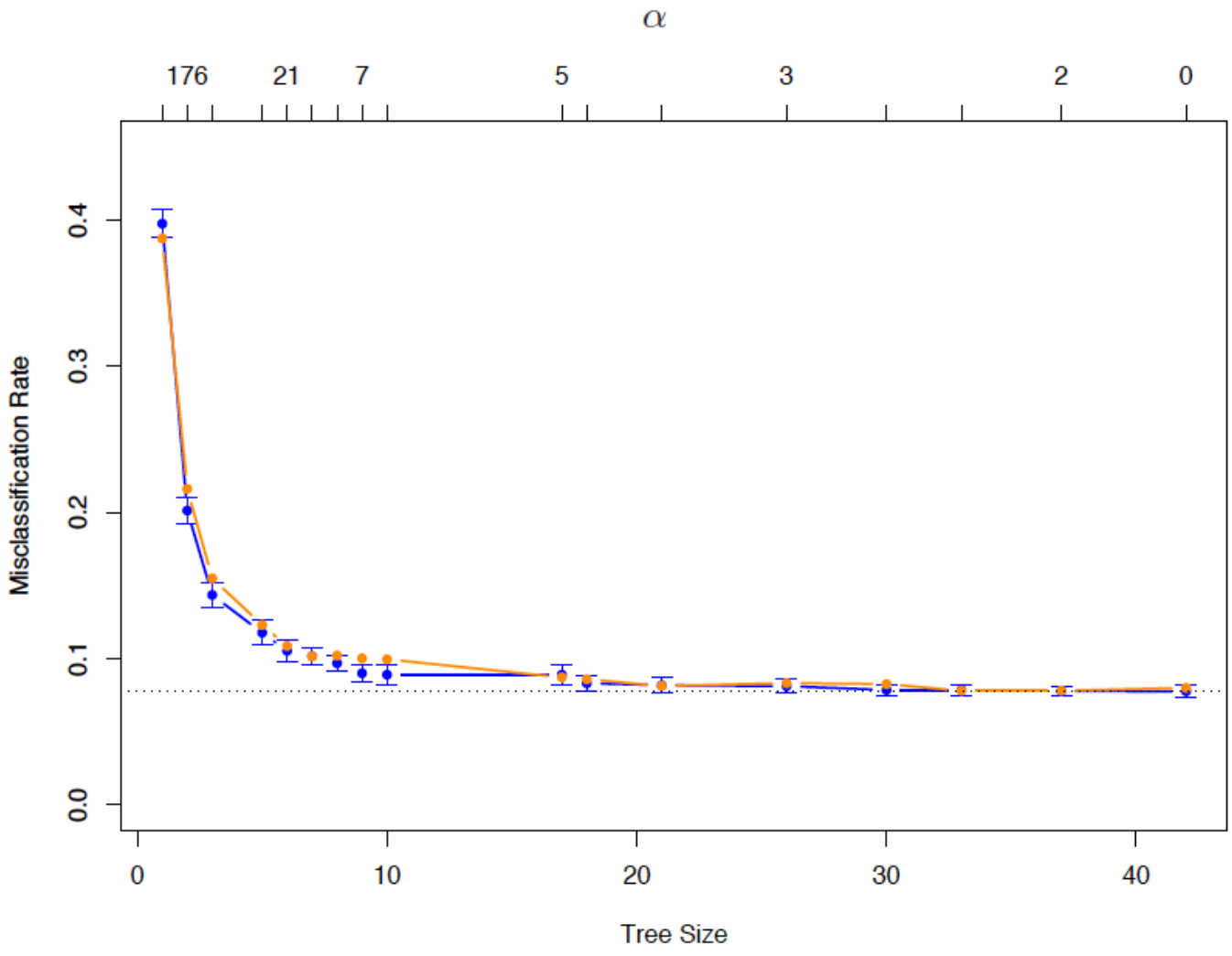
Example: $n = 4601$ emails, of which 1813 are considered spam. For each email we have $p = 58$ attributes.

The first 54 features measure the frequencies of 54 key words or characters (e.g., "free", "need", "$"). The last 3 measure

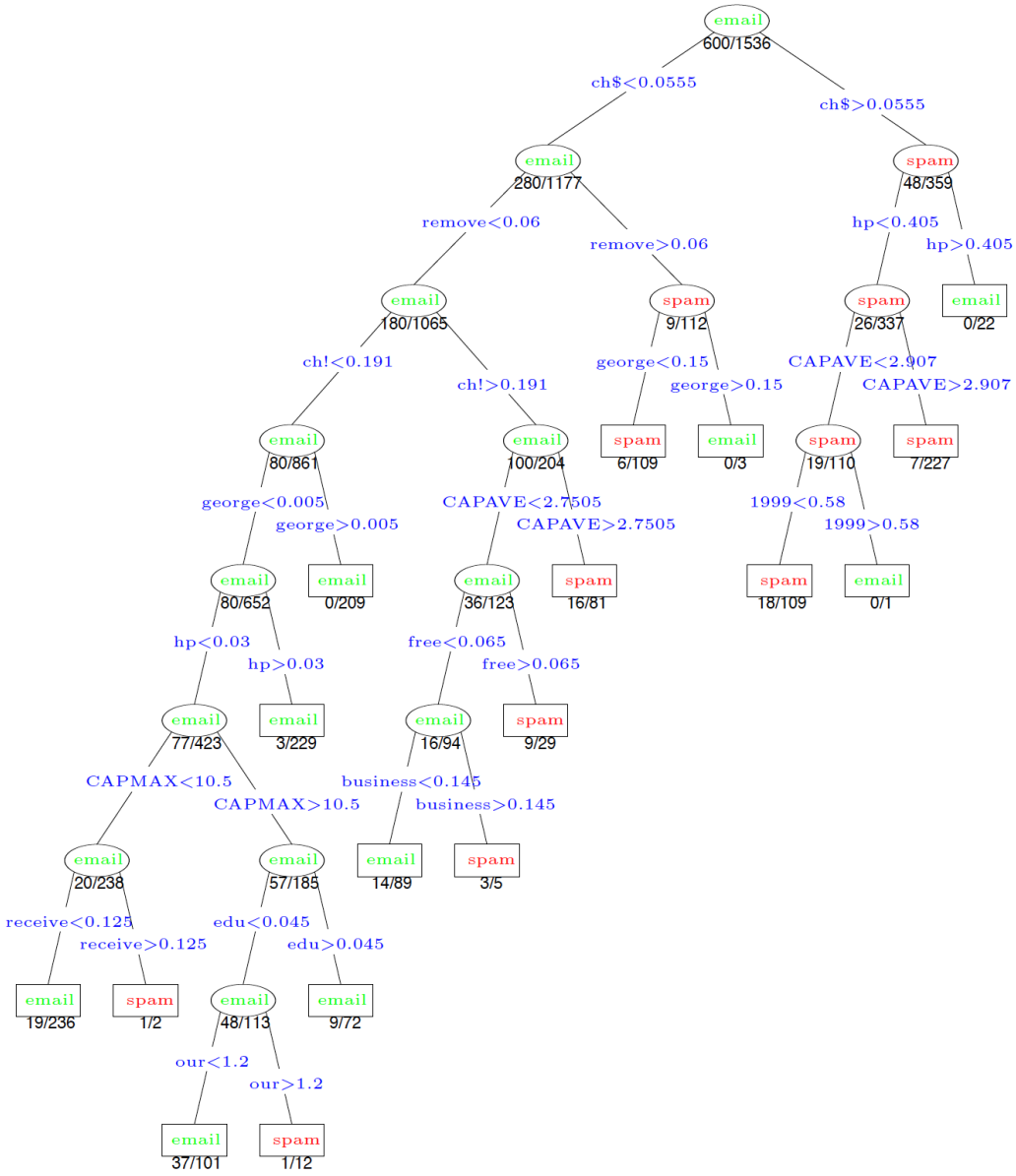
- the average length of uninterrupted sequences of capitals;
- the length of the longest uninterrupted sequence of capitals;
- the sum of lengths of uninterrupted sequences of capitals

(Data from ESL section 9.2.5)

Cross-validation error curve for the spam data (from ESL page 314):



Tree of size 17, chosen by cross-validation (from ESL page 315):



Other impurity measures

We used misclassification error as a measure of the impurity of region R_j,

$$1 - \hat{p}_{c_j}(R_j)$$

But there are other useful measures too: the Gini index:

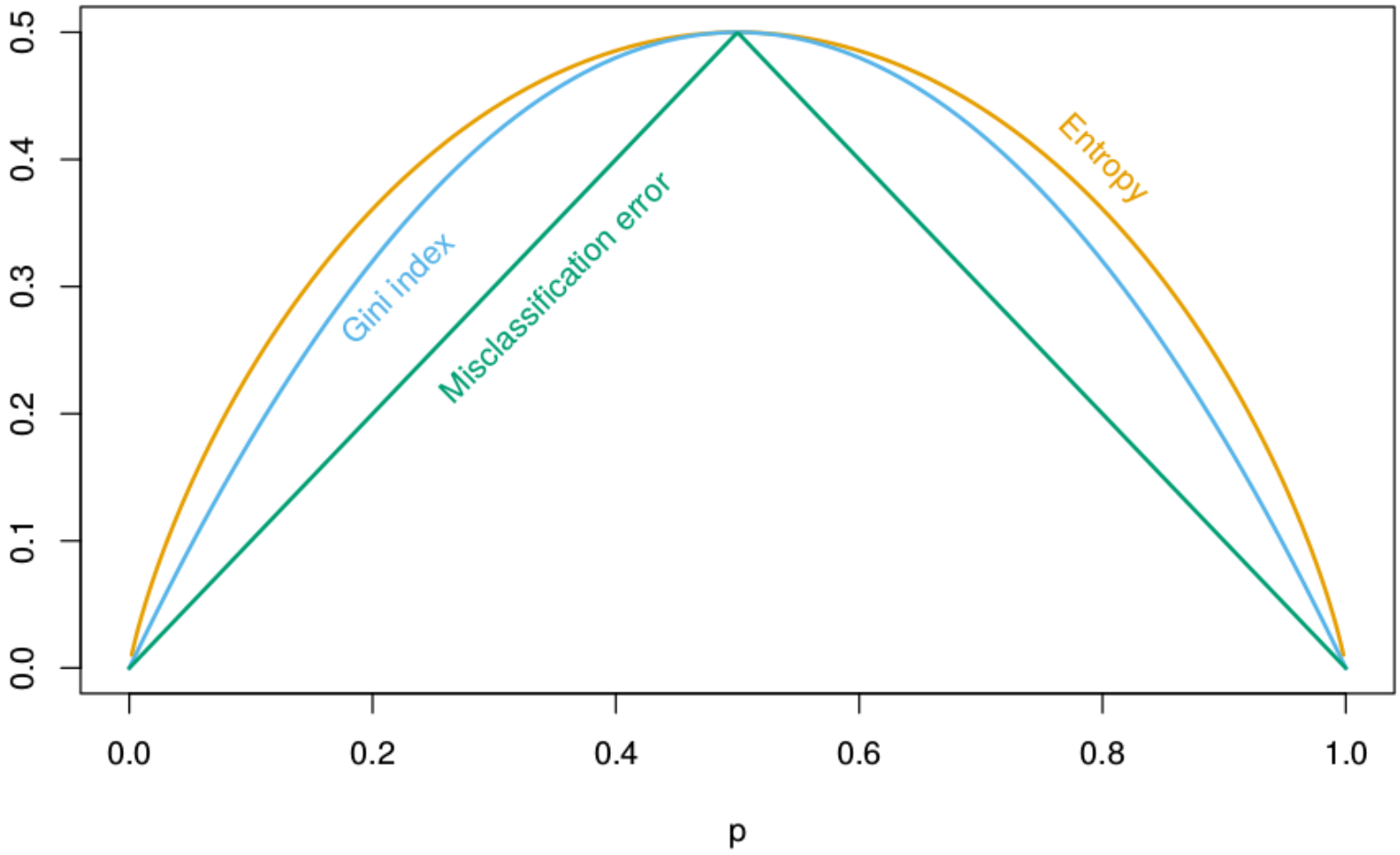
$$\sum_{k=1}^{K} \hat{p}_k(R_j)\left[1 - \hat{p}_k(R_j)\right],$$

and the cross-entropy or deviance:

$$-\sum_{k=1}^{K} \hat{p}_k(R_j) \log\left\{\hat{p}_k(R_j)\right\}.$$

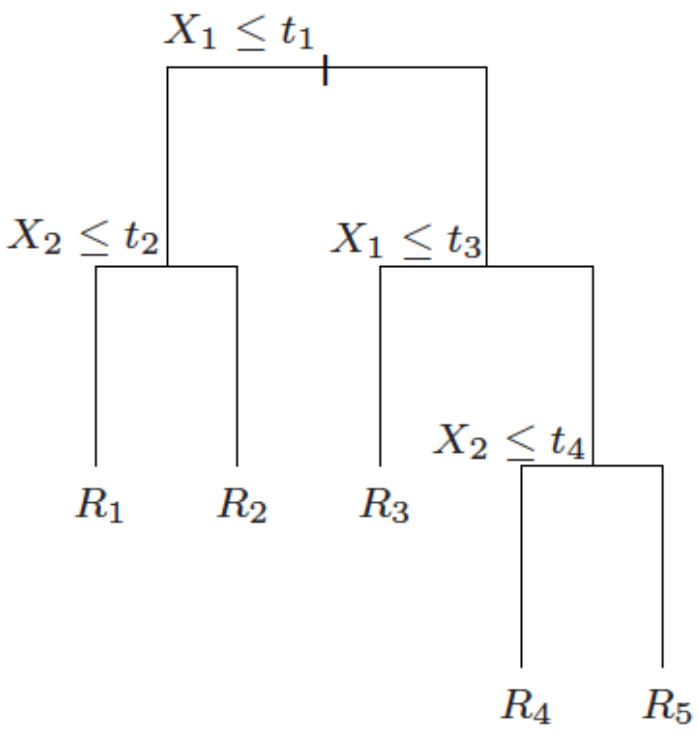
Using these measures instead of misclassification error is often better because it leads to purer nodes early on, improving the greedy search.

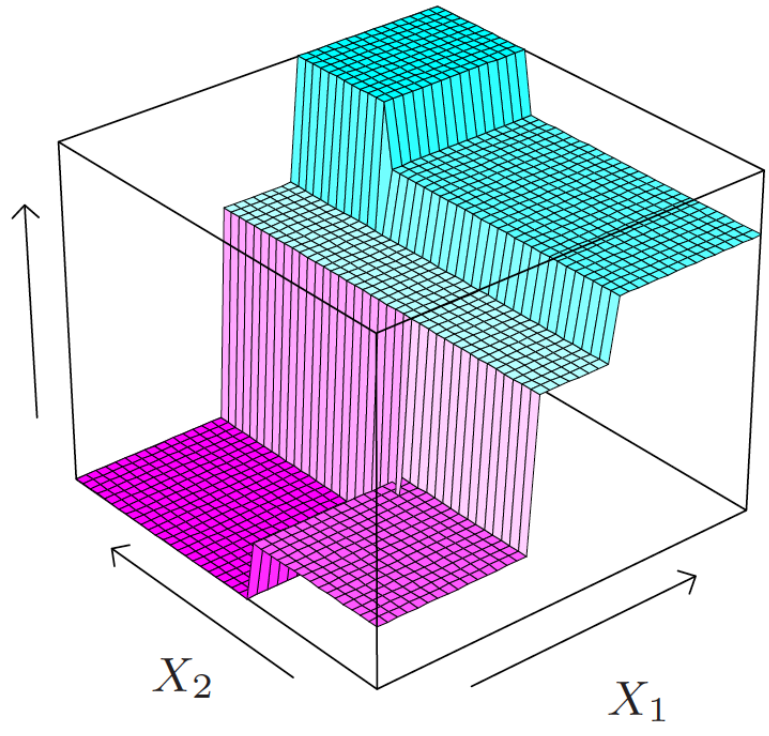
Other impurity measures



Regression trees

Suppose that now we want to predict a continuous outcome instead of a class label. Essentially, everything follows as before, but now we just fit a constant inside each rectangle





The estimated regression function has the form

$$\hat{f}^{\text{tree}}(x) \;=\; \sum_{j=1}^{m} c_j \cdot 1\{x \in R_j\} \;=\; c_j \text{ such that } x \in R_j$$

just as it did with classification. The quantities c_j are no longer predicted classes, but instead they are real numbers: the average response within each region:

$$c_j = \frac{1}{n_j} \sum_{x_i \in R_j} y_i$$

The main difference in building the tree is that we use squared error loss instead of misclassification error (or Gini index or deviance) to decide which region to split.

Categorical predictors

Awesome properties!

Trees have some amazing properties, especially when compared to other methods we've looked at.

- The ability to tune bias/variance!
- Interpretability!
- Automatic handling of variable transformations!
- Automatic detection of interactions!

How well do trees predict?

Trees have so many amazing properties! Unfortunately. . . they don't usually predict well. . .

Trees tend to suffer from high variance for a given amount of flexibility because they are quite unstable: a smaller change in the observed data can lead to a dramatically different sequence of splits, and hence a different prediction.

This instability comes from their hierarchical nature; once a split is made, it is permanent and can never be "unmade" further down in the tree

Salvaging Trees

Trees have many incredibly useful properties, but don't usually make great predictors.

Instead, we will use trees as building blocks in the construction of more complex, powerful predictive methods.

Our goal is to preserve as many of the useful tree properties as possible while dramatically improving our predictive ability.